Laboratory Guide 6: Quantum Key Distribution — The BB84 Protocol

Measurement as a Security Primitive

Course: Quantum Computing Foundation

1 Introduction and Objectives

In Lab 5 you learned that measuring a qubit in the *wrong* basis gives a random outcome and **irreversibly disturbs the state**. That looked like a limitation. In 1984, Bennett and Brassard turned it into a feature: if an eavesdropper cannot measure without disturbing, then **disturbance is evidence of eavesdropping** — and two parties can build a shared secret key whose security rests on physics, not on computational hardness.

This laboratory develops the BB84 protocol completely: the four signal states, the exact measurement statistics, sifting, the quantum bit error rate (QBER) with and without an eavesdropper, and the probability of catching her. Every number derived here is verified live in the lab's interactive simulator and rebuilt with real quantum circuits in the companion Qiskit notebook.

Prerequisites:

- Lab 5: *Quantum Measurements* — the Born rule, measurement Procedures 1 and 2, and the states $|0\rangle, |1\rangle, |+\rangle, |-\rangle$ on the Bloch sphere.
- *Quantum Communication* (lecture series).
- Familiarity with Dirac notation: $|\psi\rangle$, $\langle\phi|$, $\langle\phi|\psi\rangle$.

Connections to Previous Labs

The Lab 5 bridge. Everything in BB84 is applied Lab 5:

- The four signal states are exactly the states $|0\rangle, |1\rangle, |+\rangle, |-\rangle$ you placed on the Bloch sphere — the poles and two equator points.
- The measurement statistics are the Born rule you derived, evaluated at those four states in the Z- and X-bases (Part B).
- The distinction between **Procedure 1** (direct basis projection) and **Procedure 2** (rotate with the canonical rotation $U_X = H$, then Z-measure) — which may have seemed academic — becomes *operationally decisive* when the eavesdropper must resend a quantum state (Part D).

Mode: Interactive Simulator + Qiskit **Credits:** 1 ECTS **Estimated time:** $\approx$ 10 h

Learning Objectives:

1. **Explain the logic of BB84 and its security foundations** — random bases, mutually unbiased states, sifting, and why $|\langle z_j|x_l\rangle|^2 = 1/2$ makes a wrong-basis measurement erase the encoded bit.

2. **Perform and analyze QKD experiments** — run the protocol in the simulator and in Qiskit, compute the sifted fraction and the QBER, and compare against the exact theoretical values.

3. **Understand the measurement–disturbance–security relationship** — derive the 25% QBER of intercept-resend, see why the eavesdropper cannot avoid it, and quantify detection: $P_{\text{det}}(k) = 1 - (3/4)^k$.

4. **Discuss the relevance for cryptographic systems** — key rates, error thresholds, and what QKD does (and does not) replace in modern cryptography.

2 Part A: The Four BB84 States

Alice encodes each random bit $a \in \{0, 1\}$ in one of two randomly chosen bases: the Z**-basis** (rectilinear) or the X**-basis** (diagonal). The encoding rule: bit 0 always maps to the $+1$ eigenstate of the basis observable, bit 1 to the -1 eigenstate.

$$|\psi(0, Z)\rangle = |0\rangle, \qquad |\psi(1, Z)\rangle = |1\rangle, \qquad |\psi(0, X)\rangle = |+\rangle, \qquad |\psi(1, X)\rangle = |-\rangle, \tag{1}$$

with $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. On the Lab 5 Bloch sphere these are the poles ($\theta = 0, \pi$) and two equator points ($\theta = \pi/2$, $\varphi = 0, \pi$). In the photon-polarization picture they correspond to vertical, horizontal, and the two diagonal polarizations.

Both bases are orthonormal, and they are **mutually unbiased**. Writing $|z_0\rangle = |0\rangle$, $|z_1\rangle = |1\rangle$, $|x_0\rangle = |+\rangle$, $|x_1\rangle = |-\rangle$:

$$|\langle z_j | x_l \rangle|^2 = \frac{1}{2} \qquad \text{for all } j, l \in \{0, 1\}. \tag{2}$$

Direct computation: $\langle 0|\pm\rangle = 1/\sqrt{2}$ and $\langle 1|\pm\rangle = \pm 1/\sqrt{2}$, so every cross-basis overlap has squared modulus $1/2$.

Theory Recap

Why this is the security foundation. A measurement in the wrong basis yields a uniformly random outcome — it extracts *zero* information about the encoded bit and (Lab 5!) destroys the original state in the process. Everything that follows is this one fact, exploited systematically.

3 Part B: Measurement Statistics — the Born Table

Applying the Born rule from Lab 5,

$$P(m \mid \psi; B) = |\langle m|\psi\rangle|^2, \tag{3}$$

to the four BB84 states in both bases gives the complete statistics of the protocol:

	Z**-measurement**		X**-measurement**	
State sent	$P(0)$	$P(1)$	$P(+)$	$P(-)$
$\|0\rangle$	1	0	1/2	1/2
$\|1\rangle$	0	1	1/2	1/2
$\|+\rangle$	1/2	1/2	1	0
$\|-\rangle$	1/2	1/2	0	1

Table 1: The BB84 Born table. Matched basis ⇒ deterministic recovery of the bit; mismatched basis ⇒ coin-flip outcome, bit erased (equation (2)).

Procedure 1 vs Procedure 2

Read this carefully — it returns in Part D. Table 1 lists *probabilities*, and these are identical whether the X-basis measurement is done by **Procedure 1** (direct projection onto $\{|+\rangle, |-\rangle\}$) or by **Procedure 2** (apply the canonical rotation $U_X = H$, then Z-measure — what real hardware does). Lab 5 established this equivalence. Bob keeps only a classical bit, so *for Bob* the choice of procedure is invisible: with Procedure 2 the Z-outcome $c \in \{0, 1\}$ maps to key bit $b = c$ because $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$.

But the two procedures leave **different post-measurement states** (Procedure 1: a basis eigenstate; Procedure 2: $|0\rangle$ or $|1\rangle$). Bob never cares. **The eavesdropper will** — she must resend a quantum state, and a post-measurement state is exactly what she has in hand. Hold that thought for Part D.

4 Part C: The Protocol, Sifting, and the Error Check

For each round $i = 1, \ldots, n$:

1. **Alice** draws a uniform random bit a_i and basis $\alpha_i \in \{Z, X\}$, prepares $|\psi(a_i, \alpha_i)\rangle$, and sends it through the quantum channel.
2. **Bob** draws his own uniform random basis $\beta_i \in \{Z, X\}$ — independently, since he cannot know α_i — and measures, recording bit b_i. (On hardware: Procedure 2 with $U_X = H$.)
3. **Basis reconciliation:** over an authenticated public channel, Alice and Bob announce their *bases* — never the bits. They keep the **sifted set** $S = \{i : \alpha_i = \beta_i\}$ and discard everything else.
4. **Error check:** they sacrifice a uniformly random subset of k sifted positions, publicly compare a_i vs b_i there, and estimate the **quantum bit error rate (QBER)**. Too many errors → abort.
5. **Key:** the remaining sifted, unchecked bits form the raw shared key.

Theory Recap

Sifted fraction. α_i and β_i are independent and uniform on two values, so

$$P(\alpha_i = \beta_i) = P(Z, Z) + P(X, X) = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}. \tag{4}$$

On average **half the rounds survive sifting**: the sifted length is binomially distributed with expectation $n/2$.

Theory Recap

QBER without eavesdropping. On an ideal channel, every sifted round is a matched-basis measurement — the deterministic cells of Table 1. Hence

$$Q_{\text{no Eve}} = P(b_i \neq a_i \mid i \in S) = 0 \quad \text{exactly.} \tag{5}$$

Not "small" — **zero**. On an ideal channel, *any* error in the sifted key is evidence that something interfered with the qubits in transit.

Important

Why sifting matters. Without step 3, mismatched-basis rounds (coin flips) would contribute errors even with no eavesdropper at all: the raw, unsifted disagreement rate is $\frac{1}{2} \cdot \frac{1}{2} = 25\%$ for a perfectly secure channel. Sifting removes exactly this noise floor, which is what makes the QBER a meaningful security signal. Verify both numbers in the simulator.

5 Part D: The Intercept-Resend Attack — QBER = 25%

Eve's simplest attack: intercept each qubit, measure it in a random basis $\gamma_i \in \{Z, X\}$, and resend the post-measurement state to Bob. She cannot do better than guess — the basis announcements happen only *after* Bob has measured. Condition on a sifted round ($\beta = \alpha$) and split on Eve's guess:

- **Case 1 — Eve guesses right** ($\gamma = \alpha$, probability $\frac{1}{2}$). Matched-basis measurement is deterministic: Eve reads $e = a$, resends the very same state, Bob recovers $b = a$. Error probability: 0.
- **Case 2 — Eve guesses wrong** ($\gamma \neq \alpha$, probability $\frac{1}{2}$). Her outcome is a coin flip, and her resent state is a γ-basis eigenstate. Bob measures it in $\alpha \neq \gamma$ — mutually unbiased again — so his outcome is uniform *regardless of what Alice sent*. Error probability: $\frac{1}{2}$.

$$Q = \underbrace{\tfrac{1}{2} \cdot 0}_{\gamma=\alpha} + \underbrace{\tfrac{1}{2} \cdot \tfrac{1}{2}}_{\gamma\neq\alpha} = \tfrac{1}{4} = 25\%. \tag{6}$$

Two refinements, both testable directly in the simulator:

- **Partial interception.** If Eve intercepts each round independently with probability p and lets the rest pass, errors arise only on intercepted rounds:

  $$Q(p) = \frac{p}{4}, \qquad Q(0) = 0, \qquad Q(1) = \tfrac{1}{4}, \tag{7}$$

  linear in p. Less interception means less information for Eve *and* less visibility — but never zero of one with the other.
- **Fixed-basis Eve.** If Eve always measures in Z instead of guessing randomly, the arithmetic changes but the answer does not: her matching rounds contribute 0, the others $\frac{1}{2}$ — still $Q = \frac{1}{4}$. Randomizing her basis neither helps nor hurts her visibility.

Procedure 1 vs Procedure 2

Where the procedure distinction becomes operational. "Resend the post-measurement state" hides a subtlety. If Eve measures by **Procedure 1**, her post-measurement state *is* the γ-basis eigenstate $|e_\gamma\rangle$ — resending it gives the 25% above. If she measures by **Procedure 2** (apply U_γ, Z-measure), her post-measurement state is the computational state $|e\rangle$, and to run the same attack she must **re-encode** it by applying $U_\gamma^\dagger$ before resending. Done correctly, the two implementations are indistinguishable — same states to Bob, same $Q = 25\%$.

But suppose she forgets the re-encoding and naively resends $|e\rangle$ as-is. Work it through with Table 1:

- Alice's Z-rounds: Eve's $\gamma = Z$ causes no error; $\gamma = X$ scrambles e to a coin flip that Bob then reads deterministically $\to$ error $\frac{1}{2}$. Contribution: $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.
- Alice's X-rounds: Bob X-measures a Z-eigenstate — coin flip *whatever* Eve's basis was $\to$ error $\frac{1}{2}$. Contribution: $\frac{1}{2}$.

$$Q_{\text{naive}} = \tfrac{1}{2} \cdot \tfrac{1}{4} + \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{3}{8} = 37.5\% \;\neq\; 25\%. \tag{8}$$

Every outcome *probability* in her measurement was identical between the two procedures — yet the attack statistics changed, because the *post-measurement states* differ and the resent state is a post-measurement state. This is the Lab 5 distinction made operational. The simulator's "naive resend" mode lets you watch the QBER converge to 37.5% instead of 25%.

6 Part E: Catching Eve — Detection Probability

Under full interception, each of the k publicly compared check bits is wrong independently with probability $\frac{1}{4}$. Eve escapes only if *all* k agree:

$$P(\text{undetected}) = \left(\tfrac{3}{4}\right)^k \qquad \Longrightarrow \qquad P_{\text{det}}(k) = 1 - \left(\tfrac{3}{4}\right)^k. \tag{9}$$

check bits k	1	5	10	20
$P_{\text{det}}(k)$	0.25	≈ 0.7627	≈ 0.9437	≈ 0.9968

Table 2: Detection probability against full intercept-resend, equation (9).

Detection is **exponentially certain** in the number of check bits. For partial interception the same argument gives

$$P_{\text{det}}(k; p) = 1 - \left(1 - \frac{p}{4}\right)^k \tag{10}$$

(e.g. $p = 0.5$, $k = 16$ gives $P_{\text{det}} \approx 0.8819$). Note the $k = 0$ limit: no check bits, no detection — the public comparison is not an optional optimization, it *is* the security test.

Important

Honest security context. What you derived here covers one specific attack. The full security statement for BB84 (Shor & Preskill, 2000) says the protocol with error correction and privacy amplification produces a secure key whenever the QBER stays below $\approx 11\%$ — against *any* attack physics allows, not just intercept-resend. Intercept-resend's 25% sits far above that threshold, which is why this attack is always catastrophic for Eve. The proof machinery is beyond this lab; the QBER-threshold *logic* — measure the error rate, abort above threshold — is exactly what you practice here.

7 Part F: Guided Activities

Simulator

The lab's interactive page (`lab6_bb84.html`) contains a BB84 simulator built for this lab. **Step-by-Step** mode shows one photon at a time — watch the basis choices, Eve's interference, and the sifted key forming. **Batch Statistics** mode runs thousands of rounds and plots the measured statistics against every formula derived above. All runs are seeded: the same seed always reproduces the same run. The companion Qiskit notebook rebuilds the same protocol with real quantum circuits:

`https://colab.research.google.com/github/LMrilo/quantum-labs-2026/blob/main/lab6/lab6_bb84.ipynb`

7.1 Activity 1: Sifting Statistics (Batch mode)

Student Activity

With Eve **off**, run $N = 100$, then $N = 1000$, then $N = 5000$ (same seed). Record the sifted fraction each time.

1. How does the deviation from $1/2$ shrink as N grows? Compare with the statistical scale $\sqrt{1/(4N)}$.
2. Confirm the QBER is *exactly* 0 in every run — not just small. Why is this exact even though the simulation is random? (Equation (5).)

7.2 Activity 2: Measuring Eve (Step-by-Step + Batch modes)

Student Activity

Turn Eve **on** (intercept-resend, $p = 1$).

1. In Step-by-Step mode, find a round where Eve guessed the wrong basis but *no error resulted*. Why is that possible? (Hint: Case 2 gives an error with probability $\frac{1}{2}$, not 1.)
2. In Batch mode with $N = 5000$, verify the QBER lands near 25%. How large is the deviation? Change the seed a few times — does it stay within a band of roughly $\pm 2\sqrt{0.25 \cdot 0.75/(N/2)}$?

7.3 Activity 3: The Naive Resend — Post-States Matter

Student Activity

Switch the Eve strategy to **naive resend** (she measures via Procedure 2 and forgets to re-encode with $U_\gamma^\dagger$). Run $N = 5000$.

1. Verify the QBER converges near 37.5%, not 25% (equation (8)).
2. Every measurement *probability* in Eve's apparatus is identical in the two strategies. What, physically, is different about what Bob receives?
3. Alice's Z-rounds and X-rounds contribute differently ($\frac{1}{4}$ vs $\frac{1}{2}$). Which rounds in the Step-by-Step log show it?

7.4 Activity 4: Detection Budget (Detection panel)

Student Activity

Suppose Alice and Bob want to catch a full intercept-resend Eve with at least 99% probability.

1. From $P_{\text{det}}(k) = 1 - (3/4)^k$, compute the smallest sufficient k. Verify with the detection panel.
2. Repeat for a lazier Eve with $p = 0.25$ (equation (10)). How many check bits do you need now? What does this say about the cost of detecting weak eavesdropping?

7.5 Activity 5: The Protocol in Qiskit

The companion notebook implements the four BB84 states and the basis-selective measurement as two short routines, then scales them to the full protocol with sifting, an intercept-resend Eve (measure-and-re-prepare), QBER statistics against the 25% theory line, and the detection probability against $1 - (3/4)^k$. Its two core routines:

```python
def prepare_bb84_state(bit: int, basis: str) -> QuantumCircuit:
    """Return a 1-qubit circuit preparing the BB84 state |psi(bit, basis)>."""
    qc = QuantumCircuit(1)
    if bit == 1:
        qc.x(0)          # |0> -> |1>
    if basis == "X":
        qc.h(0)          # |0> -> |+>,  |1> -> |->
    return qc

def measure_in_basis(qc: QuantumCircuit, basis: str) -> QuantumCircuit:
    """Append a basis-B measurement (Procedure 2) to a 1-qubit circuit, in place
    ."""
    if basis == "X":
        qc.h(0)          # canonical rotation U_X = H
    qc.measure_all()
    return qc
```

The X-basis measurement is Procedure 2 with the canonical rotation $U_X = H$ — hardware has no other option. The notebook closes with an optional run on real IBM hardware, where you will meet a nonzero QBER *without* any Eve, and confront the question real QKD systems face: how do you tell noise from eavesdropping?

Important

Qiskit bit-ordering note. Qiskit orders multi-qubit bitstrings little-endian (rightmost bit = qubit q_0), while this lab series writes states big-endian (qubit 0 leftmost). In this lab every quantum register holds a *single* qubit, so the two conventions coincide and no conversion is needed — stated here for consistency with the multi-qubit labs, where the distinction is essential.

7.6 Activity 6: Quick Check

Reflection Questions

Question 1: Alice sends $|+\rangle$ and Bob measures in the Z-basis. What is the probability he reads 0?

(a) 1 (b) 1/2 (c) 0

Question 2: Why does the sifted key contain *no* errors on an ideal channel without Eve?

(a) Because errors are corrected during basis reconciliation.

(b) Because matched-basis measurements reproduce the encoded bit with probability 1.

(c) Because Bob's random basis averages the errors away.

Question 3: Eve measures a qubit in the X-basis via Procedure 2 (H, then Z-measure) and gets outcome 0. Which state does she hold afterwards?

(a) $|+\rangle$ (b) $|0\rangle$ (c) Whatever Alice sent.

8 Resources

- Bennett, C. H. & Brassard, G., "Quantum cryptography: Public key distribution and coin tossing," *Proc. IEEE Int. Conf. on Computers, Systems and Signal Processing*, Bangalore, 1984, pp. 175–179. Reprinted: *Theor. Comput. Sci.* **560**, 7–11 (2014). — The BB84 protocol.
- Nielsen, M. A. & Chuang, I. L., *Quantum Computation and Quantum Information*, Cambridge University Press, 10th anniversary edition, 2010, §12.6 (quantum cryptography: BB84, intercept-resend, security sketch).
- Gisin, N., Ribordy, G., Tittel, W. & Zbinden, H., "Quantum cryptography," *Rev. Mod. Phys.* **74**, 145 (2002). — Survey; QBER terminology and practical attack context.
- Shor, P. W. & Preskill, J., "Simple proof of security of the BB84 quantum key distribution protocol," *Phys. Rev. Lett.* **85**, 441 (2000). — The $\approx 11\%$ threshold cited in Part E.
- Lab 5: *Quantum Measurements* (this series) — Born rule, Procedures 1 and 2, the states $|0\rangle, |1\rangle, |+\rangle, |-\rangle$.
- Companion Jupyter notebook: `quantum-labs-2026/lab6/lab6_bb84.ipynb` (Colab link in Part F).